Supplementary Material S1

Stochastic Morphological π-Mixture Kernels: Identifiability, Topological Stability, and Output-Space Learning

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Reproducibility guide | Package version 2.1.0 | Prepared 6 September 2026

Purpose and scope

This archive supplies the complete computational record accompanying the manuscript: frozen inputs, fitted models, candidate outputs, primary and diagnostic results, figure and table generators, and numerical checks of the mathematical claims. The written proofs remain in the main manuscript.

The original campaign evaluates a fixed morphological bank on a held-out test set. The subsequent output-space construction reuses that inspected set and is explicitly exploratory. Reproducing these calculations does not create a new independent validation.

Primary experiment at a glance

Dataset	Mask size	Train / validation	Scored test
Shapes	48 × 48	1,000 / 300	499
MNIST	28 × 28	2,000 / 500	1,000
Fashion-MNIST	28 × 28	2,000 / 500	1,000

The 31-action bank is frozen before the final primary evaluation. Six primary corruption conditions produce 14,994 noisy cases; two additional conditions produce 4,998 diagnostic cases. Five fitted seeds (101-105) are retained for each dataset, together with three learned local-filter models.

Mean primary composite loss

Dataset	Validation-best fixed filter	Conditional selector
shapes	0.044996	0.040324
mnist	0.044702	0.042749
fashion	0.153080	0.130340

Lower is better. These are image-cluster means across primary corruptions and fitted seeds. The comparisons concern this fixed bank and study design, not all denoising methods. Full metric tables and paired intervals are in results/.

Protocol and indexing

Frozen data and exclusions

MNIST and Fashion-MNIST retain selected original pixels, thresholded masks (intensity ≥ 128), and original partition row IDs. Training and validation come from official training partitions; evaluation uses official test partitions. Development-pilot IDs and identical binary masks are quarantined. The shapes file retains 500 prospective test masks; ID 173 duplicates validation and is excluded from scoring, leaving 499.

Corruption schedule

Index / name	Deletion	Addition	Role
0 balanced04 / 1 balanced10	.04 / .10	.04 / .10	Primary
2 salt04 / 3 salt10	.01 / .01	.04 / .10	Primary
4 pepper04 / 5 pepper10	.04 / .10	.01 / .01	Primary
6 shift20 / 7 clean	.20 / 0	.20 / 0	Diagnostic

Training and validation use one corruption per image. Each final test image uses all eight conditions. The exact corruption law and seeded random-number construction are in run_campaign.py. Repeated corruptions of the same image are clustered for reporting.

Loss, topology, and uncertainty

The objective is 0.4(1-Dice) + 0.2(binary MSE) + 0.2 times each capped, target-normalized component and hole error. Separately reported Betti-error columns contain raw absolute errors. Foreground uses 8-connectivity; background uses 4-connectivity with a permanent exterior frame. β_0 counts foreground components; β_1 counts bounded background components.

Sampled-method metrics are exact expectations over the finite categorical law. Mean-mask methods are scored separately. Aggregate PSNR is $-10\log_{10}$(mean binary MSE). The bootstrap uses 2,000 clean-image cluster resamples; six paired control comparisons use Bonferroni-adjusted percentile intervals. These descriptive intervals do not certify performance on a new noise distribution.

Decode and join arrays correctly

Bank scores have axes case $\times$ action $\times$ metric. Metric order is loss, MSE, Dice, IoU, component error, hole error, exact topology. Action order is recorded in environment.json. Map image_index through the clean split IDs to recover a public source row. Primary CSVs use condition names; quotient CSVs use numeric condition indices.

Candidate masks are packed on the width axis in big bit order: unpack with count=W. Noisy masks are flattened before packing: unpack with count=$H \times W$ and then reshape. Detailed field names and commands are in docs/REPRODUCIBILITY.md.

Reproduce the evidence

Environment and integrity

Use Python 3.12 and install the pinned requirements in a fresh virtual environment. No GPU or source-data download is needed for ordinary reproduction. The archive includes the clean masks and selected public pixels. The wrapper sets one thread for BLAS and OpenMP; fitted forests use n_jobs$=1$.

```
python -m pip install -r requirements.txt
python code/check_package.py
```

The checksum check verifies every frozen release file. Run it before making edits. The reproduction wrapper copies the package to a new directory and rejects an existing destination. All generated files and timings are written into that copy.

Three execution levels

```
python code/reproduce.py --level quick --workdir ../smm-quick
python code/reproduce.py --level analysis --workdir ../smm-analysis
python code/reproduce.py --level full --workdir ../smm-full
```

Level	Executed work
quick	Original mathematical checks; regenerated masks/features for two cases per dataset; all 15 forests replayed on those cases; frozen-pair invariance checked.
analysis	Quick checks, full saved-model replays, seed-101 fresh refits, all summaries/figures, expanded mathematical checks, quotient validation, and this guide.
full	Recompute all candidate banks from frozen clean masks, refit all models, then perform the analysis level.

Where each result comes from

summarize.py regenerates primary metrics, bootstrap intervals, ablations, diagnostic plots, qualitative examples, and gradient-variance diagnostics. validate_kernel_quotient.py regenerates the exploratory output-space table and saved probabilities. verify_release.py checks splits, reconstructed scores, all saved forest predictions, fresh seed-101 refits, and CPU timings.

audit_math.py and audit_kernel_theory.py contain finite mathematical checks; plot_topology_sharpness.py generates the explicit extremal masks. docs/RESULT_MAP.md links the manuscript evidence to exact filenames. models/ contains every fitted estimator. data/source_manifest.json preserves source archive URLs and hashes.

Expected agreement

Discrete actions and masks should agree under the pinned stack. Saved float32 arrays introduce rounding: replay probability tolerances are 10^{-7} and sampled score recomputation tolerances are 2×10^{-6}. Timings, PDF metadata, and compressed-container bytes can differ. Compare scientific values rather than regeneration timestamps.

Kernel validation and limits

Exploratory output-space comparison

The output law uses the minimum predicted action risk within each exact-output class and a uniform reference over distinct outputs. Copying an existing frozen output/score pair preserves that law. Refitting after changing the feature representation is outside the invariance claim. Temperature is selected from .005, .02, and .1 using the original validation set; all 15 models select .005.

Dataset	Mean distinct outputs	Action Gibbs loss	Output-space loss
shapes	26.724	0.041460	0.041698
mnist	23.545	0.043322	0.043493
fashion	26.236	0.131275	0.131510

The output-space loss is slightly higher on all three datasets. Its demonstrated benefit here is invariance to redundant action representation. There are 14,880 direct replication checks on retained cases; the largest residual is below 7×10^{-16}. This extension is not an independent confirmatory benchmark.

Numerical support for the proofs

Independent graph calculations compare 66,146 masks and 1,053,698 single-pixel changes, check 2,304 exact local component identities, and include separate constructions attaining both sharp sensitivity bounds. The suite also runs 2,000 entropy/risk/replication trials, 32 finite transport comparisons, Bayesian-reversal checks, and gradient comparisons.

The largest finite-difference gradient residual in the expanded suite is 8.2×10^{-11}. These finite and numerical checks complement the quantified written proofs; they are not a proof-assistant certificate or external mathematical review.

Limits to retain in any presentation

Individual metrics do not all improve: MNIST Dice and shapes hole error can worsen. Stronger Fashion-MNIST corruption gives a documented failure against the fixed area filter. Clean inputs can change. Evaluating a full 31-action bank is materially slower than using a single area filter. Low-resolution binary masks and simulated corruptions do not establish RGB, clinical, real-sensor, or high-resolution performance.

The smooth morphology control is not a full implementation of BiMoNN, DMNN, or SoftMorph. No superiority over those published architectures is claimed. The mathematical contraction statements require their stated coefficient conditions; the fitted model has not been shown to satisfy a global contraction coefficient below one.

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